

Reflection Principle

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1 Brownian bridge

Setup

- log-price X_t follows BM $dX = \mu' dt + \sigma dW$
 - Where $\mu' = \mu - \frac{1}{2}\sigma^2$ and $dS = S(\mu dt + \sigma dW)$
- $M = \sup_{0 \leq t \leq T} X_t$.

Useful equations

- $P(M(t_1) = m | X(t_0) = x_0) \sim \text{Rayleigh}(\sigma\sqrt{t_1 - t_0})$
- $P(M(t_1) = m | X(t_0) = x_0, X(t_1) = x_1) = \exp\left(-\frac{2(m-x_0)(m-x_1)}{\sigma^2(t_1-t_0)}\right)$
 - Reflection principle can be used to prove this.

Thinking about $\sigma = 1, \mu' = 0$ is enough. It is because Girsanov theorem ensures that the process is still a Brownian motion under dirftless measure.

It is convenient that the Girsanov factor cancels out.

$$\begin{aligned} & P^{\mathbb{P}}(M(t_1) = m | X(t_0) = x_0, X(t_1) = x_1) \\ &= \frac{P^{\mathbb{P}}(M(t_1) = m, X(t_1) = x_1 | X(t_0) = x_0)}{P^{\mathbb{P}}(X(t_1) = x_1 | X(t_0) = x_0)} \\ &= \frac{P^{\mathbb{P}}(M(t_1) = m, X(t_1) = x_1 | \mathcal{F}_{t_0})}{P^{\mathbb{P}}(X(t_1) = x_1 | \mathcal{F}_{t_0})} \\ &= \frac{P^{\mathbb{Q}}(M(t_1) = m, X(t_1) = x_1 | \mathcal{F}_{t_0}) \frac{d\mathbb{P}}{d\mathbb{Q}}}{P^{\mathbb{Q}}(X(t_1) = x_1 | \mathcal{F}_{t_0}) \frac{d\mathbb{P}}{d\mathbb{Q}}} \\ &= \frac{P^{\mathbb{Q}}(M(t_1) = m, X(t_1) = x_1 | \mathcal{F}_{t_0})}{P^{\mathbb{Q}}(X(t_1) = x_1 | \mathcal{F}_{t_0})} \\ &= P^{\mathbb{Q}}(M(t_1) = m | X(t_0) = x_0, X(t_1) = x_1) \end{aligned}$$

2 Reflection Principle

Let us consider $dX = dW$.

The following figure shows two distribution

- Solid line (left) : $X(t)$, $X(t_0) = x_0$, $X(t_1) = x_1$
- Dashed line (right) : $X^*(t)$, $X^*(t_0) = 2m - x_0$, $X^*(t_1) = 2m - x_1$

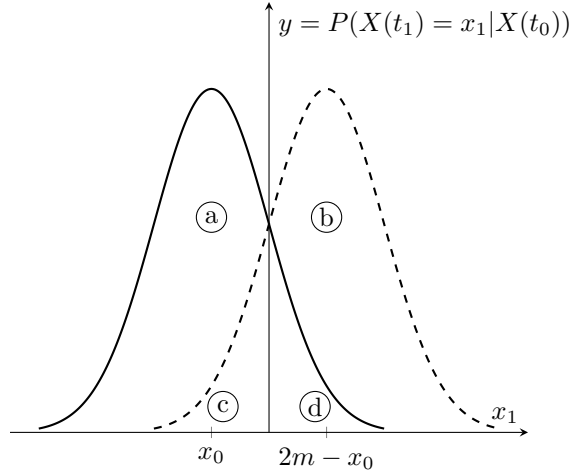


Figure 1: $X^*(t)$ (dashed line) is a mirror image of $X(t)$ (solid line).

(a) and (d) represents area under $X(t)$ (solid line)
(c) and (b) represents area under $X^*(t)$ (dashed line)

- When there is no bouncy wall placed at m
 - (a) : $P(X(t_1) \leq m | X(t_0) = x_0, X(t_1) = x_1)$,
 - (d) : $P(X(t_1) \geq m | X(t_0) = x_0, X(t_1) = x_1)$
- When there is a bouncy wall placed at m
 - (a) : $P(1 | X(t_0) = x_0, X(t_1) = x_1)$
 - **(a)-(c) = (a)-(d)** : $P(M(t_1) < m | X(t_0) = x_0, X(t_1) = x_1)$
 - (c) : $P(M(t_1) = m | X(t_0) = x_0, X(t_1) = x_1)$ (touched wall)

It makes sense because, starting from x_0 and hitting the wall is equivalent to starting from $2m - x_0$ and crossing the wall.

Be careful, the point can cross(bounce) the wall multiple times.
However, it does not matter because due to the symmetry, we can flip the cases that crosses the line odd times.